

CRYPTOGRAPHIC RELEASE RECORD ARCHITECTURE & RUNTIME ENFORCEMENT PIPELINE

Immutable Engineering Ledger $\longleftrightarrow$ Hardware Root of Trust $\longleftrightarrow$ Envelope Trichotomy $\longleftrightarrow$ Automatic Invalidation Triggers

1. IMMUTABLE RELEASE RECORD LEDGER

Claim Identifier

CLAIM-CH16-TACTILE-001

GSN Tree Topology Hash

SHA256: 8f3c4e...d94a

Verified Evidence Pointer Array

- LOG-TAC-2026-0814 (P99.9: 12 ms)
- DYN-IMP-2026-0922 (JF dt: 1.1 N-s)
- LOG-MCU-2026-1105 (Jitter: 5 μ s)

Cryptographic Manifest Hashes

- Policy Checkpoint: SHA256: e4b7...10c2
- Safety FPGA Bitstream: SHA256: 3a91...fe78
- FreeRTOS Kernel: SHA256: 7d18...99b0

Standing Conditions Contract

1. Max Payload Mass: $m \leq 1.5$ kg
2. Optical Feeder Dryness: $< 2\%$ sheen
3. Torque Observer Trip: $F_{trip} = 18$ N

Signed Adjudicator Attestation

Lead Systems Adjudicator Key:
Ed25519: sig_44b9_88fa... (Verified)
Expiry: 2000 h / 10^6 cycles

Stored in Machine Notebook Ledger (Section 4.10)

2. HARDWARE ROOT OF TRUST & ENFORCER

Hardware Secure Boot & Manifest Verification

- Public key burned into silicon OTP eFuses
- Dual-Bank (A/B) flash with watchdog fallback
- 1-bit hash mismatch halts inverter gate power

Refusal to energize without signed release record

Runtime Envelope Trichotomy (Section 16.5)

1. KNOWN-TRUE (Inside Envelope)

Telemetry valid $\cdot$ Covariance bounded $\rightarrow$ Permit $u_t = p_t$

2. KNOWN-FALSE (Physical Boundary Breached)

$F_{norm} > 18$ N or tracking error > 50 mm $\rightarrow$ Trip brakes < 1 ms

3. UNKNOWN (Epistemic Gap / Loss of Telemetry)

CRC fail / blind lidar $\rightarrow$ MUST TREAT AS OUTSIDE $\rightarrow$ E-Stop

1 kHz Synchronous Condition Detectors

- Continuously Monitored: Load cell payload mass check ($m \leq 1.5$ kg) before every arm acceleration cycle
- Optical Dryness Strobe: Pre-grasp surface reflectance
- Motor Thermistors: Winding temperature $T \leq 393$ K
- Periodically Re-established: Joint torque calibration (500 h)
- Friction Brake Inspection: Lining wear check (250k cycles)

3. INVALIDATION TRIGGERS & RE-DECISION

Software Invalidation Triggers

- Any policy neural network weight update
- RTOS thread scheduling priority change
- Lower-level sensor driver / CAN patch
 $\rightarrow$ Automatically voids SHA-256 signature

Hardware Invalidation Triggers

- Unverified actuator or sensor part swap
- Gearbox backlash $> 0.10^\circ$ (15 ms phase lag)
- Brake lining wear < 2.0 mm thickness
 $\rightarrow$ Hardware interlock inhibits power stage

Operational Invalidation Triggers

- Ambient temperature $T > 40^\circ\text{C}$
- DC Bus voltage droop < 21.6 V (10% drop)
- Workpiece mass > 1.5 kg
 $\rightarrow$ Immediate safe stop & operational latch

Continuous Re-Decision Protocol

- Non-volatile circular buffer logs all residuals
- Periodic empirical audit every 500 op hours
- Sensor noise drift $> 15\%$ triggers derating
- Re-adjudication required to renew authority

REVOCATION OF OPERATING AUTHORITY

Inverter gates clamped $\cdot$ Mandatory re-audit